

AI Resume Screening System

Using LangChain, LangSmith & Groq (LLaMA 3)

Project	AgentialAI Internship – Innomatics Research Labs
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Date	April 16, 2026
Technology Stack	Python · LangChain · LangSmith · Groq API · PyPDF

1. Introduction

In today's recruitment process, manually screening resumes is time-consuming and inefficient. Organizations receive hundreds of applications for a single role, making it difficult for recruiters to identify suitable candidates quickly. This project presents an AI-powered Resume Screening System that automates the evaluation process using Large Language Models (LLMs). The system analyzes resumes against a job description and provides a fit score along with an explanation, simulating a real-world recruiter workflow.

2. Problem Statement

The goal is to build an AI tool for recruiters with the following functionality:

- **Input:** Resume + Job Description
- **Process:** Skill Extraction → Matching → Scoring
- **Output:** Fit Score (0–100) + Explanation

The system should:

- Extract skills, tools, and experience
- Compare candidate profile with job requirements
- Assign a score
- Provide explainable reasoning

3. Objectives

- Build a modular LLM-based pipeline using LangChain
- Implement resume and job description parsing
- Develop skill matching and scoring logic
- Generate explainable outputs
- Enable debugging and tracing using LangSmith

4. Tools and Technologies Used

Technology	Purpose
Python	Core programming language
LangChain	Pipeline orchestration
LangSmith	Tracing and debugging

Groq API (LLaMA 3)	LLM model inference
PyPDF	PDF resume parsing

5. System Architecture

The system follows a modular pipeline:

Resume → Extract → JD Extract → Match → Score → Explain

Pipeline Steps:

- 1. **Extract:** Extract skills, tools, and experience from resume
- 2. **JD Extract:** Extract required skills from job description
- 3. **Match:** Compare candidate skills with required skills
- 4. **Score:** Calculate a score between 0–100
- 5. **Explain:** Provide reasoning behind the score

6. Implementation Details

Step 1 – Skill Extraction

The extraction chain reads the PDF resume using PyPDF and prompts the LLM to return structured JSON containing:

- Skills (Python, SQL, ML tools, etc.)
- Tools (Docker, Git, etc.)
- Experience in years

Step 2 – Matching Logic

The match chain receives candidate skills and required skills and returns:

- **Matched skills** = intersection of candidate and required skills
- **Missing skills** = required skills absent from candidate profile

Step 3 – Scoring Logic

The scoring formula is:

$$\text{Score} = (\text{Matched Skills} / \text{Total Required Skills}) \times 100$$

Adjustments applied:

- +10 if candidate has ≥ 3 years of experience
- -10 if candidate has < 1 year of experience

Step 4 – Explanation

The explain chain generates a human-readable breakdown covering matched skills, missing critical skills, and experience impact.

7. Results

Three candidates with different profiles were evaluated against a Data Scientist / AI Engineer job description:

Candidate	Profile Type	Score	Key Observation
Ashish Saval	Strong	73.68	14/20 skills matched; missing Matplotlib, OpenAI API, HuggingFace, G
Nitin Satarkar	Average	10.0	Only 2/20 skills matched (Python, SQL); 0.5 yrs experience
Ram Kachare	Weak / Mismatch	5.0	1/20 skill matched (REST APIs); primarily a web/Java developer

Observations: Strong candidates scored higher due to relevant ML/AI skills. Weak candidates lacked required technical skills. Profile mismatches (e.g., a Java/web developer) showed minimal alignment with AI/ML job requirements.

8. LangSmith Tracing & Screenshots

The following screenshots were captured during the development and testing of the pipeline.

8.1 Average Candidate – Nitin Satarkar (Score: 10.0)

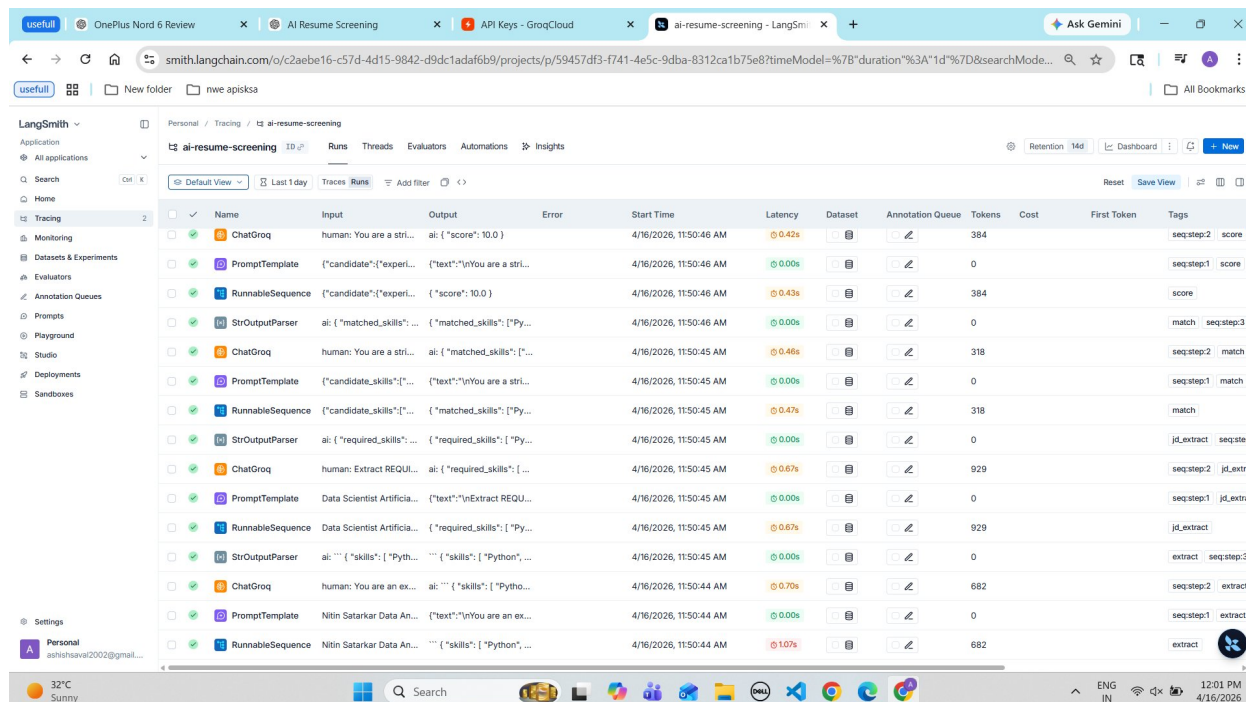


Fig 1 – LangSmith Runs overview showing all pipeline steps for the average candidate

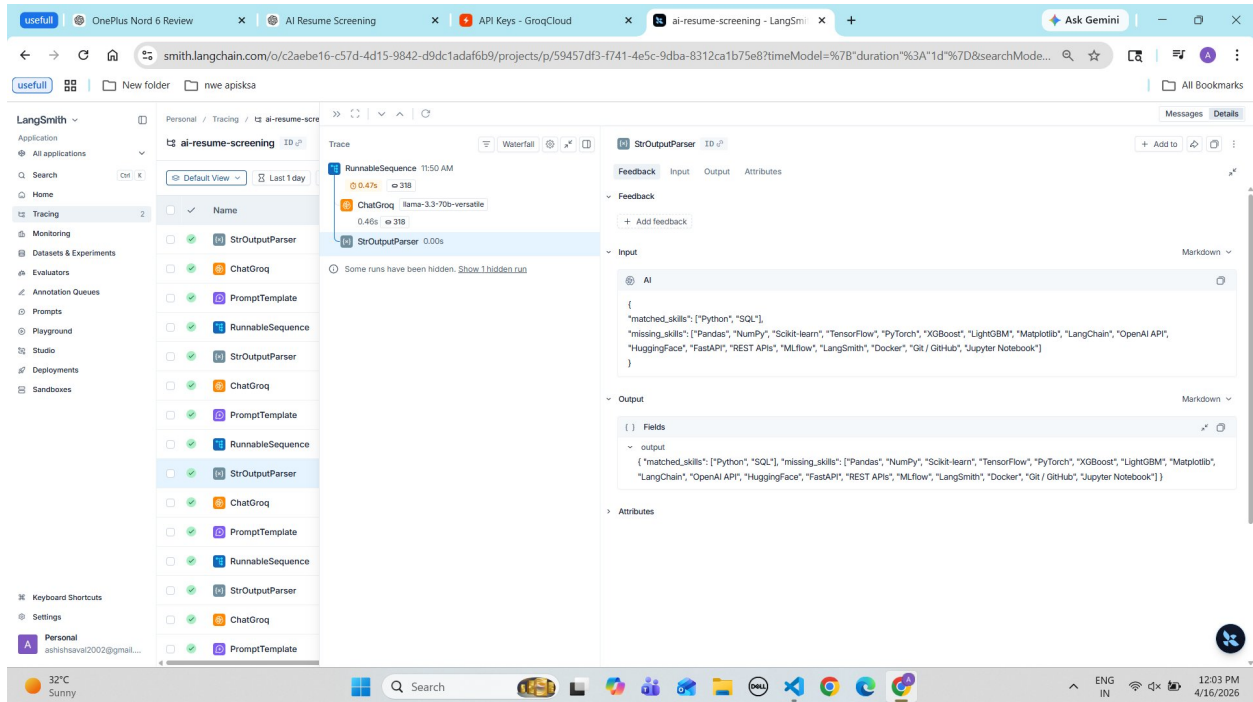


Fig 2 – StrOutputParser trace: matched skills and missing skills JSON output

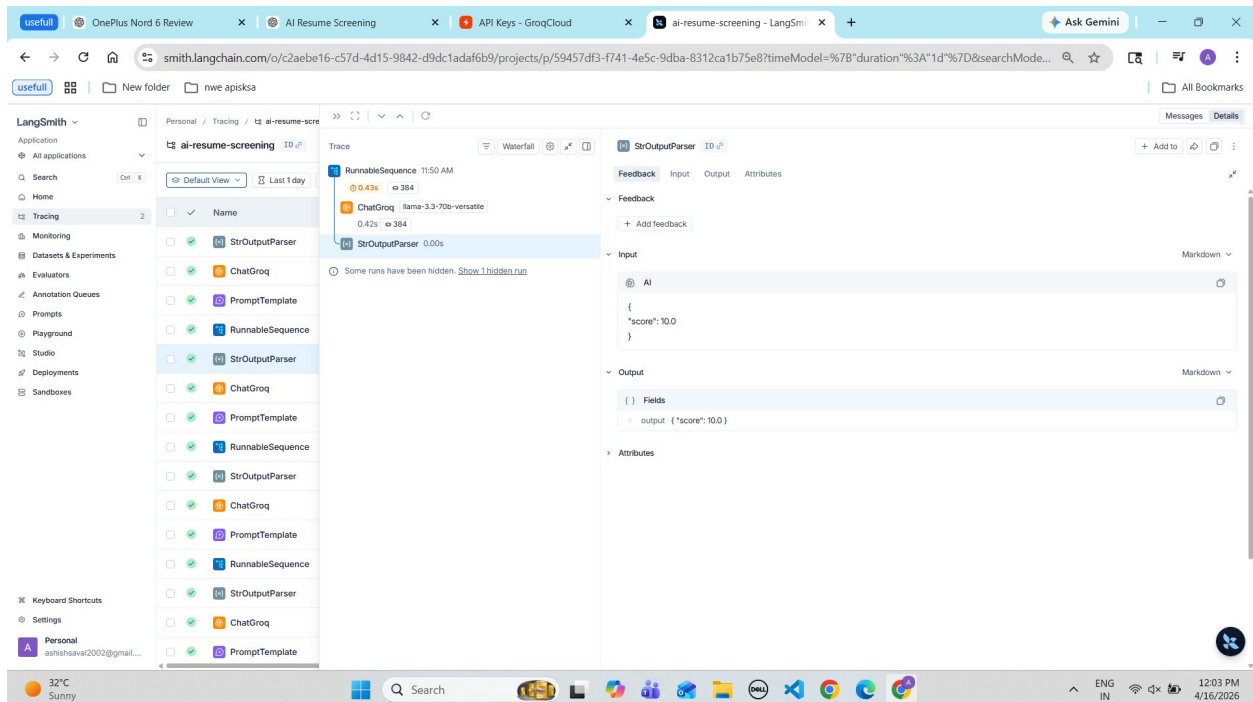


Fig 3 – Score output: { score: 10.0 } after evaluation

8.2 Strong Candidate – Ashish Saval (Score: 73.68)

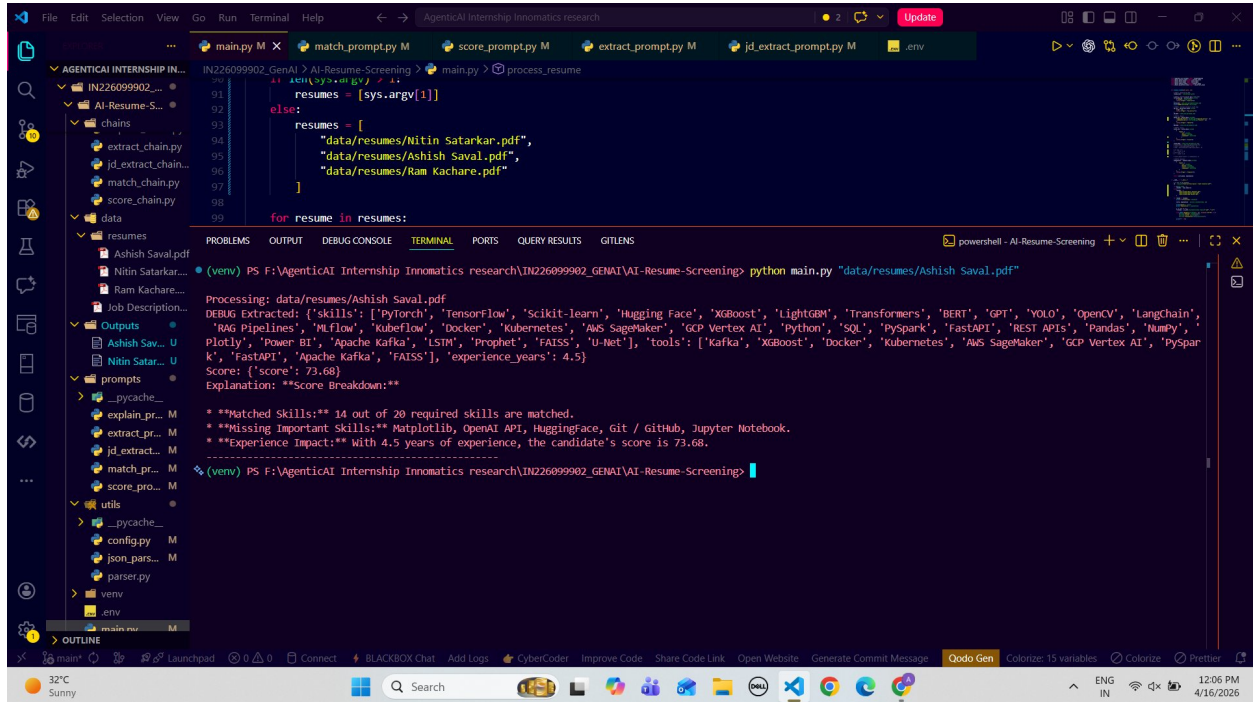


Fig 4 – LangSmith Runs: full trace for strong candidate pipeline execution

Name	Input	Output	Error	Start Time	Latency	Dataset	Annotation Queue	Tokens	Cost	First Token	Tags
PromptTemplate	{ "candidate": { "experience_years": 4.5, "skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 } }	{ "text": "You are a string of text." }		4/16/2026, 12:06:24 PM	0.00s			0			score seq:step:1
RunnableSequence	{ "candidate": { "experience_years": 4.5, "skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 } }	{ "score": 73.68 }		4/16/2026, 12:06:24 PM	0.47s			536			score
StrOutputParser	ai: { "matched_skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }	ai: { "matched_skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }		4/16/2026, 12:06:23 PM	0.00s			0			match seq:step:3
ChatGroq	human: You are a string of text.	ai: { "matched_skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }		4/16/2026, 12:06:23 PM	0.71s			438			match seq:step:2
PromptTemplate	{ "candidate_skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }	{ "text": "You are a string of text." }		4/16/2026, 12:06:23 PM	0.00s			0			match seq:step:1
RunnableSequence	{ "candidate_skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }	{ "matched_skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }		4/16/2026, 12:06:23 PM	0.71s			438			match
StrOutputParser	ai: { "required_skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }	ai: { "required_skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }		4/16/2026, 12:06:23 PM	0.00s			0			jd_extract seq:step:1
ChatGroq	human: Extract REQUI...	ai: { "required_skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }		4/16/2026, 12:06:22 PM	0.82s			929			jd_extract seq:step:1
PromptTemplate	Data Scientist Artificial Intelligence Engineer	{ "text": "Extract REQUI..." }		4/16/2026, 12:06:22 PM	0.00s			0			jd_extract seq:step:1
RunnableSequence	Data Scientist Artificial Intelligence Engineer	{ "required_skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }		4/16/2026, 12:06:22 PM	0.83s			929			jd_extract
StrOutputParser	ai: { "skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }	ai: { "skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }		4/16/2026, 12:06:22 PM	0.00s			0			extract seq:step:3
ChatGroq	human: You are an ex...	ai: { "skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }		4/16/2026, 12:06:21 PM	1.21s			1,140			seq:step:2 extract
PromptTemplate	Ashish Saval AI / ML Engineer	{ "text": "You are an ex..." }		4/16/2026, 12:06:21 PM	0.00s			0			seq:step:1 extract
RunnableSequence	Ashish Saval AI / ML Engineer	{ "skills": ["PyTorch", "TensorFlow", "Scikit-learn", "Hugging Face", "XGBoost", "LightGBM", "Transformers", "BERT", "GPT", "YOLO", "OpenCV", "LangChain", "RAG Pipelines", "MLflow", "Kubeflow", "Docker", "Kubernetes", "AWS SageMaker", "GCP Vertex AI", "Python", "SQL", "DySpark", "FastAPI", "REST APIs", "Pandas", "NumPy", "Plotly", "Power BI", "Apache Kafka", "LSTM", "Prophet", "FAISS", "U-Net"] }, "tools": ["kafka", "XGBoost", "docker", "kubernetes", "AWS SageMaker", "GCP Vertex AI", "DySpark", "FastAPI", "Apache Kafka", "FAISS"], "experience_years": 4.5 }		4/16/2026, 12:06:21 PM	1.75s			1,140			extract

Fig 5 – Detailed match trace showing matched and missing skills

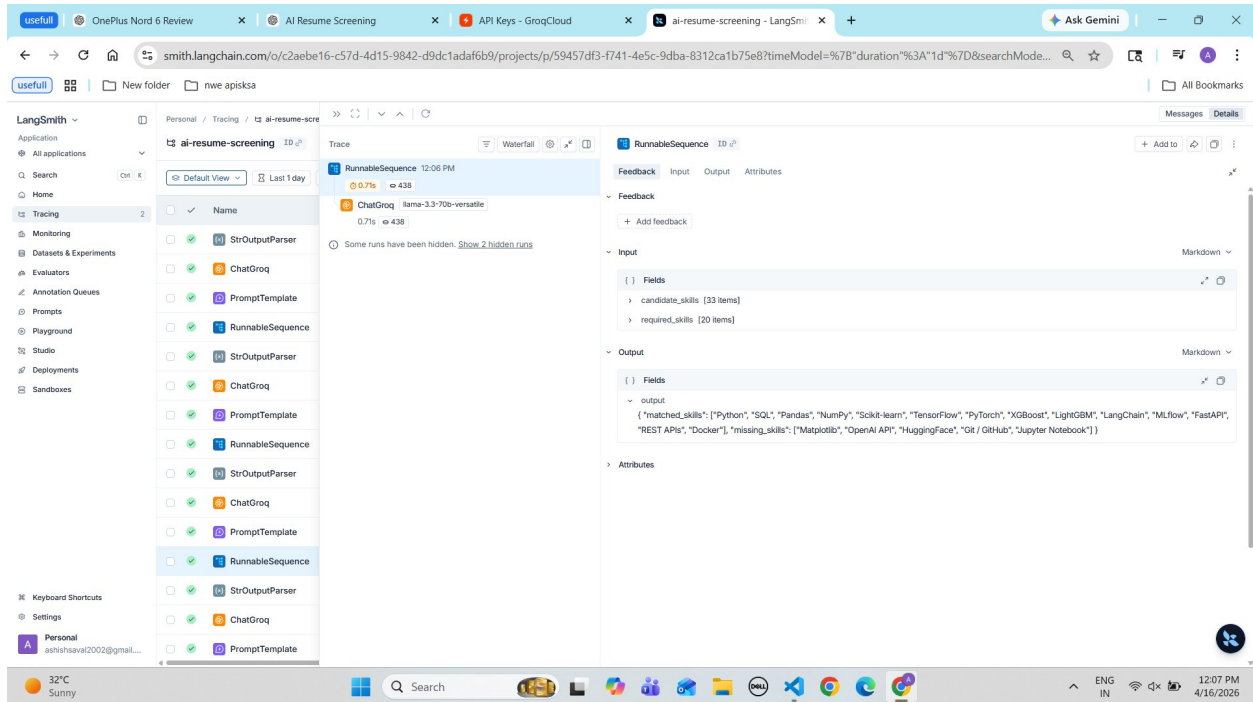


Fig 6 – RunnableSequence showing 33 candidate skills vs 20 required skills

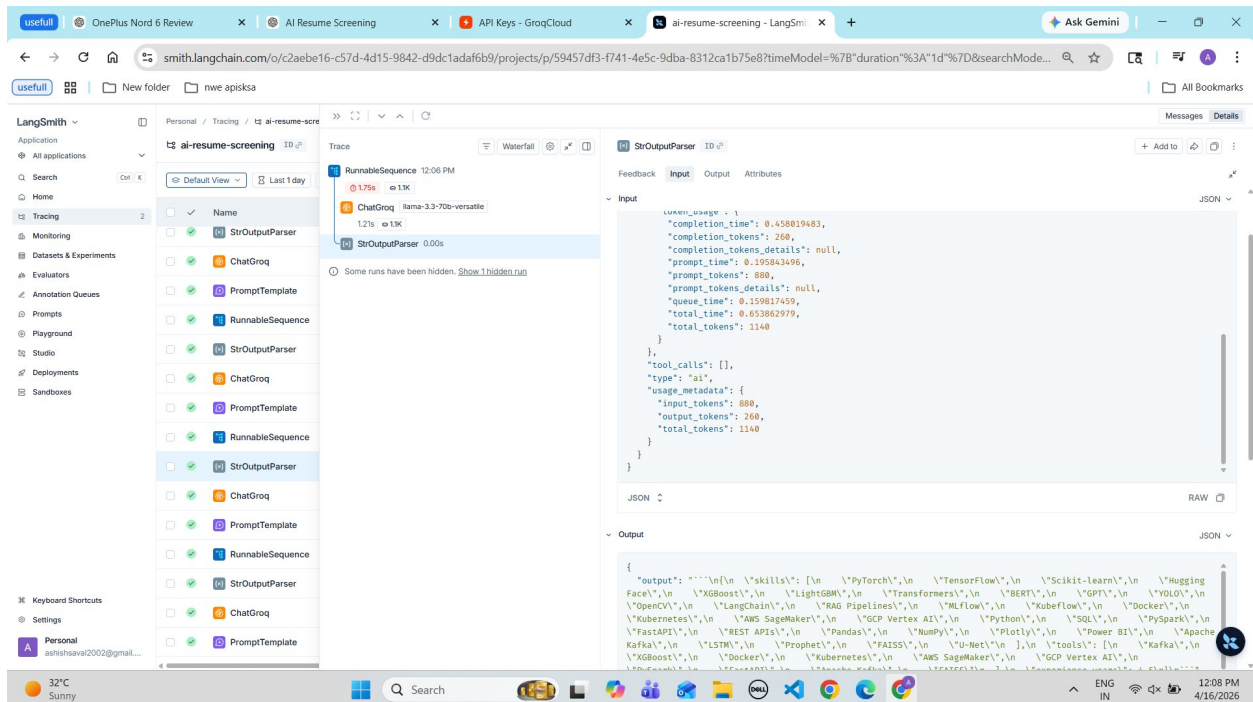
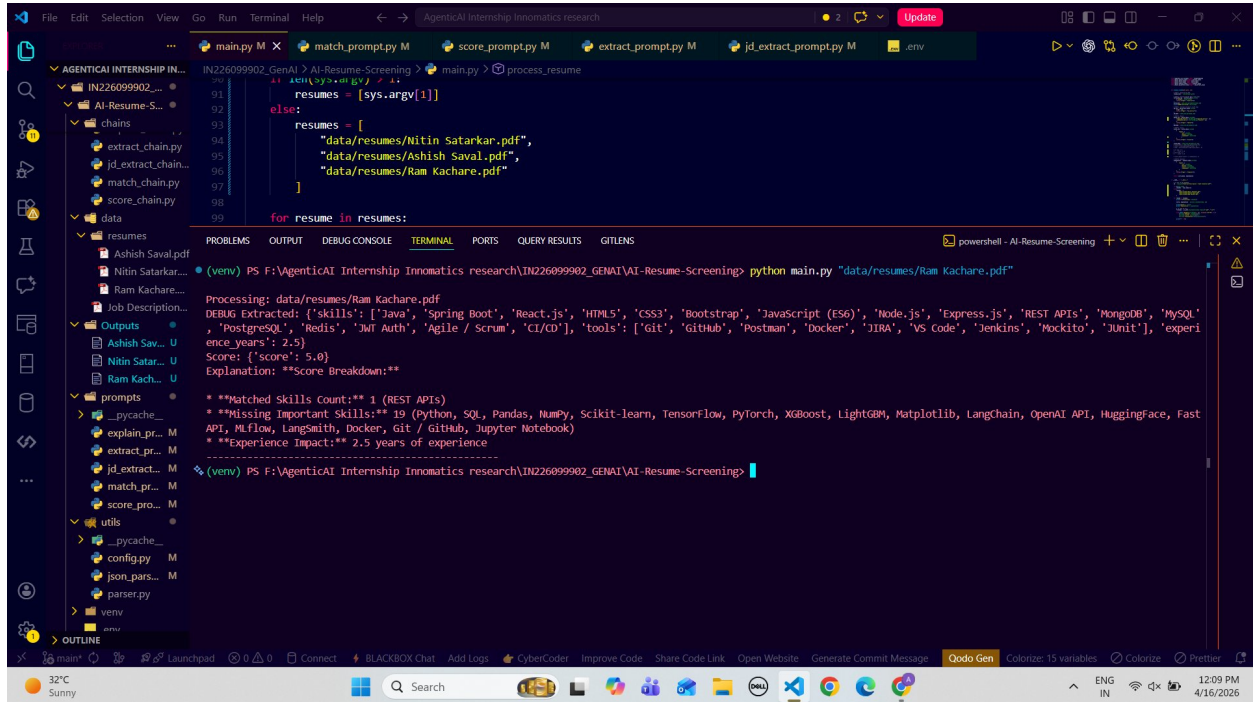


Fig 7 – Raw token usage and input/output JSON from the extraction chain

8.3 Weak / Mismatch Candidate – Ram Kachare (Score: 5.0)

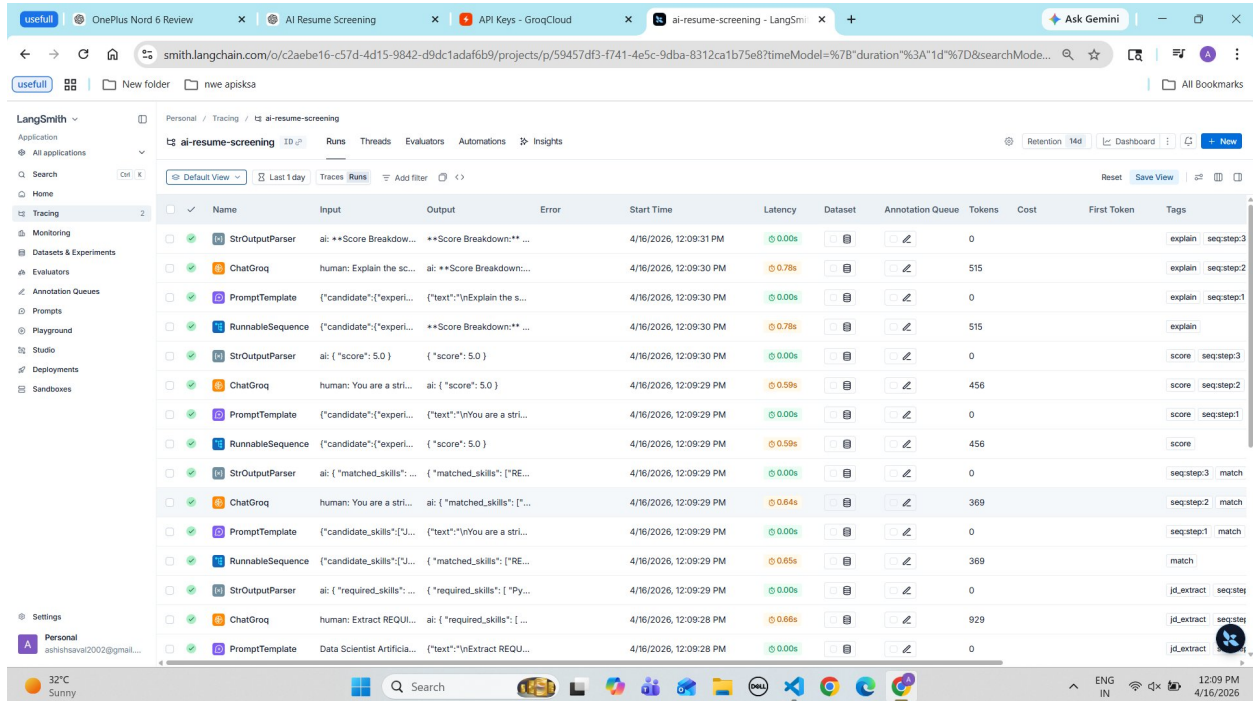


```
Processing: data/resumes/Ram Kachare.pdf
DEBUG extracted: {'skills': ['Java', 'Spring Boot', 'React.js', 'HTML5', 'CSS3', 'Bootstrap', 'JavaScript (ES6)', 'Node.js', 'Express.js', 'REST APIs', 'MongoDB', 'MySQL', 'PostgreSQL', 'Redis', 'JWT Auth', 'Agile / Scrum', 'CI/CD'], 'tools': ['Git', 'Github', 'Postman', 'Docker', 'JIRA', 'VS Code', 'Jenkins', 'Mockito', 'JUnit'], 'experience_years': 2.5}
Score: {'score': 5.0}
Explanation: **Score Breakdown:**

**Matched Skills Count:** 1 (REST APIs)
**Missing Important Skills:** 19 (Python, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Matplotlib, LangChain, OpenAI API, HuggingFace, FastAPI, MLflow, LangSmith, Docker, Git / Github, Jupyter Notebook)
**Experience Impact:** 2.5 years of experience

(venv) PS F:\AgenticAI Internship Innomatics research\IN226099902_GENAI\AI-Resume-Screening> python main.py "data/resumes/Ram Kachare.pdf"
```

Fig 8 – VS Code terminal: RAM Kachare processing output showing web skills extracted



Name	Input	Output	Error	Start Time	Latency	Dataset	Annotation Queue	Tokens	Cost	First Token	Tags
StrOutputParser	ai: **Score Breakdown...	**Score Breakdown:** ...		4/16/2026, 12:09:31 PM	0.00s			0			explain seq:step:3
ChatGroq	human: Explain the sc...	ai: **Score Breakdown...		4/16/2026, 12:09:30 PM	0.78s			515			explain seq:step:2
PromptTemplate	("candidate":("experi...	("text":("nExplain the s...		4/16/2026, 12:09:30 PM	0.00s			0			explain seq:step:1
RunnableSequence	("candidate":("experi...	**Score Breakdown:** ...		4/16/2026, 12:09:30 PM	0.78s			515			explain
StrOutputParser	ai: ("score": 5.0)	("score": 5.0)		4/16/2026, 12:09:30 PM	0.00s			0			score seq:step:3
ChatGroq	human: You are a stri...	ai: ("score": 5.0)		4/16/2026, 12:09:29 PM	0.59s			456			score seq:step:2
PromptTemplate	("candidate":("experi...	("text":("nYou are a stri...		4/16/2026, 12:09:29 PM	0.00s			0			score seq:step:1
RunnableSequence	("candidate":("experi...	("score": 5.0)		4/16/2026, 12:09:29 PM	0.59s			456			score
StrOutputParser	ai: ("matched_skills": ...	("matched_skills": ("RE...		4/16/2026, 12:09:29 PM	0.00s			0			seq:step:3 match
ChatGroq	human: You are a stri...	ai: ("matched_skills": ("...		4/16/2026, 12:09:29 PM	0.64s			369			seq:step:2 match
PromptTemplate	("candidate_skills":("...	("text":("nYou are a stri...		4/16/2026, 12:09:29 PM	0.00s			0			seq:step:1 match
RunnableSequence	("candidate_skills":("...	("matched_skills": ("RE...		4/16/2026, 12:09:29 PM	0.65s			369			match
StrOutputParser	ai: ("required_skills": ...	("required_skills": ("Py...		4/16/2026, 12:09:29 PM	0.00s			0			jd_extract seq:step:3
ChatGroq	human: Extract REQU...	ai: ("required_skills": ("...		4/16/2026, 12:09:28 PM	0.66s			929			jd_extract seq:step:2
PromptTemplate	Data Scientist Artific...	("text":("nExtract REQU...		4/16/2026, 12:09:28 PM	0.00s			0			jd_extract seq:step:1

Fig 9 – LangSmith Runs list for weak candidate showing explain, score, match steps



9. LangSmith Tracing

LangSmith was integrated as the primary observability layer for the pipeline:

- Tracking each pipeline step (extract, jd_extract, match, score, explain)
- Debugging errors such as invalid JSON parsing issues
- Monitoring input-output flow across all LLM chains
- Viewing token usage, latency, and model metadata per run

Key Benefits:

- Transparency in LLM execution
- Easy debugging with step-by-step trace inspection
- Waterfall view to identify bottlenecks and latency

10. Debugging Case

During development, an issue occurred where the model returned invalid JSON output, causing parsing errors in the StrOutputParser step.

Solution Applied:

- Enforced STRICT JSON output instructions in all prompts
- Used structured prompt templates to avoid model hallucination
- Added JSON validation logic before downstream chain steps

These changes significantly improved system reliability and output consistency.

11. Key Features

- Modular LangChain pipeline design (each step is an independent chain)
- Structured JSON outputs enforced via prompt engineering
- Explainable AI results with score breakdown
- Automated output file generation per candidate
- LangSmith tracing integration for full observability

12. Conclusion

This project demonstrates how LLMs can be used to build real-world AI systems for recruitment. The system successfully automates resume screening with scoring and explanation, providing:

- Faster candidate evaluation compared to manual review
- Transparent decision-making through explainable AI outputs
- Scalable solution that can handle multiple candidates and job roles

13. Future Enhancements

- Build a UI using Streamlit or React for easy recruiter interaction
- Add a weighted scoring system prioritizing critical skills
- Integrate a database for storing and querying results
- Support multiple job roles and dynamic JD templates

14. Learning Outcomes

- Designed and implemented LLM-based agentic pipelines
- Applied prompt engineering techniques for structured output
- Built explainable AI systems with traceable decisions
- Used LangSmith for production-grade debugging and tracing